

The Route Is the Reason: Clinical Concept Routing for Faithful 3D Medical Segmentation

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Abstract. Deep segmentation models now rival expert accuracy on brain-tumor benchmarks, yet clinical adoption lags because their decisions cannot be *faithfully* explained. The dominant remedy—post-hoc attribution (Grad-CAM, gradients, integrated gradients, occlusion)—computes a saliency map *after* the prediction; even when such a map scores well on input-sensitivity tests, it reports *where* the model is sensitive, not *which clinical concept* a region is, and it is reconstructed from the output rather than read from the decision. We propose **Clinical Concept Routing (CCR)**, a different principle: replace the encoder bottleneck with a routing layer that assigns every image token to one of K clinically-labeled expert networks. The routing probability distribution is, simultaneously, the model’s clinical concept assignment, the segmentation mechanism, and a per-voxel explanation—computed in a single forward pass, with no post-hoc step. CCR is a drop-in bottleneck module; we instantiate it as **CCR-Net** (a Swin encoder with a routing bottleneck and a UNETR decoder), detail its router, expert bank, dispatch mechanism, five-term objective, and three-phase curriculum, and characterize the sense in which its explanation is structurally faithful, with alignment quality measured by the Concept Alignment Score (CAS). We argue that a clinical explanation must satisfy two requirements at once: it must be *concept-labeled*, and it must be *distinct from and causally upstream of* the prediction. On the BraTS 2024 glioma benchmark these requirements separate three families. Post-hoc saliency (Grad-CAM, gradients, integrated gradients, occlusion) is upstream but cannot discriminate the clinical concepts (AUROC 0.52–0.73). The model’s own segmentation posterior discriminates them almost perfectly (0.98) but *is* the prediction—it restates the output rather than exposing what produced it, and its decoder divergence is zero by construction. CCR routing satisfies both: it discriminates concepts at 0.88, far above every post-hoc method, while remaining upstream ($DD > 0$) and intrinsic. We report the full battery openly, including where CCR loses: integrated gradients and occlusion win deletion *sensitivity*, which they optimize directly, and the posterior wins discriminability. Two controls show the remaining gap is largely structural rather than informational—matching the posterior to the router’s 16^3 decision budget removes about half of it, and on edema the two become statistically indistinguishable—while CCR *exceeds* the posterior on insertion. Ablations confirm concept alignment is *learned* (it collapses without the alignment loss) and that warm-up is a stabi-

lizing default, not a requirement. Dropped *unchanged* into a SegResNet CNN (CCR-SegResNet), the same module reproduces every result from scratch (CAS_{fig} up to 0.96), so the principle is backbone-agnostic.

Keywords: Interpretability · Faithfulness · Medical image segmentation · Mixture-of-experts · Brain tumor

1 Introduction

Brain-tumor segmentation models now match or exceed expert radiologists on benchmark data, but clinical deployment has not followed. The barrier is not accuracy; it is the absence of a *trustworthy, verifiable* account of what the model computed and why. A radiologist asked to act on a segmentation needs to know which clinical concept the model assigned to each region and to trust that this account reflects the actual computation.

The dominant response is post-hoc attribution: after the model predicts, apply Grad-CAM [14], integrated gradients [16], SHAP [9], or attention roll-out [1] to reconstruct a saliency map that *approximates* the decision mechanism. The structural problem is that these methods were designed for single-scalar-output models; in dense prediction every voxel has its own output, and the extension proceeds through heuristics—averaging gradients, summing attention weights, approximating Shapley values over spatial regions—each introducing a gap between the attribution and the actual computation. More fundamentally, the explanation is built from *byproducts* of computation (gradients, attention weights), not from the computation itself, and it is a scalar *sensitivity* map, not a concept assignment. A strong sensitivity method can even score well on input-sensitivity tests such as the deletion/insertion game [11]—efficiently ranking the high-leverage voxels—but a high sensitivity score does not make a map a clinical explanation: it still cannot say *which* concept a region is, and it is reconstructed from the output rather than being the decision. The right response is to make the decision itself interpretable [13].

We propose a different principle. **Clinical Concept Routing (CCR)**: in an encoder-decoder segmentation model, replace the bottleneck with a routing layer that assigns each image token to one of K clinically-labeled expert networks {Background, Necrotic core, Edema, Enhancing tumor}. The routing distribution, computed at the bottleneck, is the model’s clinical concept assignment for each token, and it *actively conditions* all downstream computation: it is causally upstream of the prediction, not derived from it. The explanation is not reconstructed from the prediction—it *is* the decision mechanism. Because the routing assignment is read directly from the forward pass, the per-voxel, concept-labeled explanation comes for free, deterministically, every time the model predicts (Fig. 1).

What a clinical explanation has to be. Stating the requirement precisely is what makes the comparison meaningful, so we adopt two criteria. An explanation

must be (i) **concept-labeled**—it must say *which* clinical concept a region is, not merely that the region matters—and (ii) **distinct from and causally upstream of the prediction**, so that reading it tells us something about how the output was produced rather than restating the output. These criteria are not rhetorical: each eliminates a different family. Post-hoc saliency satisfies (ii) but fails (i)—a sensitivity map has no concept semantics, and empirically the strongest post-hoc method separates the BraTS concepts at AUROC 0.73 while integrated gradients and Grad-CAM sit near chance. The model’s own segmentation posterior satisfies (i) trivially and fails (ii) completely: it discriminates concepts at 0.98 *because it is the concept prediction*, offering no account of what produced it. This is not a straw baseline—it is free from the same forward pass, and any evaluation of concept explanations that omits it is incomplete—but an explanation that is identical to the thing being explained is vacuous. CCR routing is, to our knowledge, the first construction for 3D dense prediction that satisfies both criteria simultaneously.

Contributions.

1. **The CCR principle and a backbone-agnostic module.** The first method to route tokens to *clinically labeled* experts at the segmentation bottleneck so that routing is at once the prediction and a per-voxel, concept-labeled explanation, for 3D dense prediction. CCR is a drop-in bottleneck module: we instantiate it, *unchanged*, on a Swin transformer (CCR-Net) and a SegResNet CNN (CCR-SegResNet), and it transfers (Sec. 3, Sec. 5.6). We detail the router, expert bank, soft/hard dispatch, five-term objective, and curriculum.
2. **A faithfulness framework.** We define the Concept Alignment Score (CAS) and Decoder Divergence (DD), and distinguish faithfulness (structural) from alignment quality (CAS) from causal relevance (deletion AUC)—three things routinely conflated in the literature (Sec. 4).
3. **An honest characterization against *both* competing families.** Under one identical protocol on all 168 test cases, we evaluate CCR against four post-hoc baselines *and* against the segmentation posterior—the intrinsic baseline that concept-explanation evaluations typically omit. CCR is the only method satisfying both criteria above; we report plainly that it is not first on any single metric, and we run two controls (a granularity-matched posterior, and insertion) that isolate how much of the residual gap is decision budget rather than information (Sec. 5).
4. **Ablations** isolating the role of the alignment loss (concept alignment is learned, not incidental) and of curriculum warm-up (the routing aligns even without it—a robustness result).

2 Related Work

Post-hoc attribution. Grad-CAM [14] weights feature maps by the gradient of the output w.r.t. that feature; integrated gradients [16] integrate gradients along a baseline path with axiomatic guarantees but no faithfulness guarantee for dense prediction; SHAP [9] approximates Shapley values, intractable in high

dimension; attention rollout [1] propagates attention through layers but conflates attention magnitude with causal contribution. All operate *after* prediction and produce an input-*sensitivity* map, not a concept assignment. On dense prediction, strong sensitivity methods (integrated gradients, occlusion) can even score well on deletion/insertion [11], yet—as we show (Sec. 5.4)—their maps do not discriminate the clinical concepts. CCR does not improve post-hoc attribution; it removes the structural reason it cannot be a concept explanation, by making the decision intrinsic [13].

Intrinsic interpretability. Concept Bottleneck Models [7] route predictions through a human-readable concept layer whose activation *is* the explanation; ProtoPNet [4] classifies by prototype similarity (“this looks like that”). Both are designed for scalar-output classification, neither handles dense prediction, and neither offers a formal faithfulness statement. CCR extends the intrinsic-interpretability principle to 3D volumetric segmentation, where the explanation must be per-voxel and concept-labeled.

Mixture-of-experts routing. Sparse MoE [15,5,12] was introduced for *efficiency*: conditional computation that processes each input with a subset of parameters. Vision and medical adaptations reuse routing for efficiency, domain adaptation, or missing-modality robustness. None of these works label experts with clinical concepts, enforce routing–concept alignment with an explicit loss, or treat the routing decision as the model’s explanation. CCR’s objective—faithful intrinsic explanation—is different in kind and requires a different design: clinically-labeled experts, an alignment loss, and bottleneck placement.

3 Method

3.1 Overview and the CCR principle

Let an encoder map a multi-modal volume $x \in \mathbb{R}^{C_{\text{in}} \times H \times W \times D}$ to a set of bottleneck tokens $\{h_t\}_{t=1}^N$, $h_t \in \mathbb{R}^d$, arranged on a coarse spatial grid. The CCR module replaces the plain bottleneck with three components: a *router* that produces, per token, a distribution over K clinically-labeled experts; an *expert bank* $\{E_k\}_{k=1}^K$; and a *dispatcher* that combines expert outputs according to the routing. The routing volume $\{p_k(t)\}$ is, at once, (i) the model’s clinical concept assignment, (ii) a quantity that conditions every downstream computation, and (iii) the explanation read out for the user. We instantiate $K=4$ to match the BraTS annotation protocol exactly: {Background, NCR, Edema, ET}.

What kind of explanation this is. CCR does not attach a post-hoc saliency map to a fixed black box; it makes the model’s decision interpretable *by construction* [13]. Because the architecture must route each region to exactly one clinically-labeled expert *before* the decoder can act, the routing is a load-bearing, concept-level decision variable—which expert processed a region, and with what

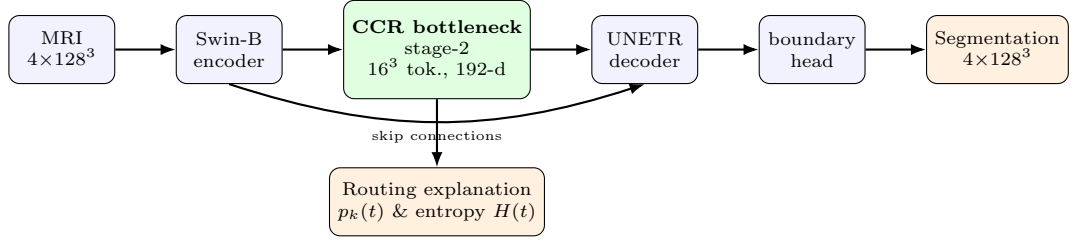


Fig. 1. CCR-Net. A Swin-B encoder produces hierarchical features; the CCR bottleneck is inserted at stage-2 (16^3 tokens, 192-d). Its expert-processed output feeds a UNETR decoder (with the usual multi-scale skips) and a thin boundary head to yield the segmentation. The same routing distribution that conditions the decoder is read out directly as the per-voxel, concept-labeled explanation and uncertainty—one forward pass, no post-hoc step.

confidence (entropy)—not a heatmap reconstructed from the output. It is causally upstream of the prediction and, as we measure (DD, Sec. 4), *not* identical to it; the per-concept experts make it modular, so one can inspect which expert fired and intervene by ablating it (the deletion analysis of Sec. 5.4). This is a different object from a monolithic segmentation posterior and from a post-hoc attribution: the explanation *is* the decision, read from the computation—so “is routing just the prediction?” misreads the claim. Routing is the concept-level decision that *produces* the prediction, exposed faithfully rather than reconstructed.

3.2 CCR-Net architecture

Figure 1 shows the full network. The encoder is a Swin-B transformer [8,6] producing five hierarchical feature maps at resolutions $64^3, 32^3, 16^3, 8^3, 4^3$ with channel dimensions 48, 96, 192, 384, 768. We insert the CCR module at *stage-2* (16^3 tokens, 192-d): this depth balances spatial resolution against semantic context. A shallower insertion gives more tokens but features too low-level to discriminate necrotic core from enhancing tumor (we observed over-routing to NCR and a ~ 0.17 drop in CAS); a deeper insertion is too coarse. The $16^3 \times 192$ feature map is flattened to a token sequence $\{h_t\}_{t=1}^{N=4096}$, processed by the CCR module (Sec. 3.3), reshaped back to 192×16^3 , and passed as the post-CCR skip connection to a MONAI UNETR decoder [3] that also receives the remaining encoder skips. A residual boundary-refinement head ($< 100k$ parameters) sharpens edges. The routing volume is trilinearly upsampled to 128^3 to form the per-voxel explanation; the decoder output is the segmentation. The CCR module depends only on a bottleneck feature grid, not on the backbone: we later insert it *unchanged* at the bottleneck of a SegResNet CNN (CCR-SegResNet, Sec. 5.6) to show the principle is backbone-agnostic.

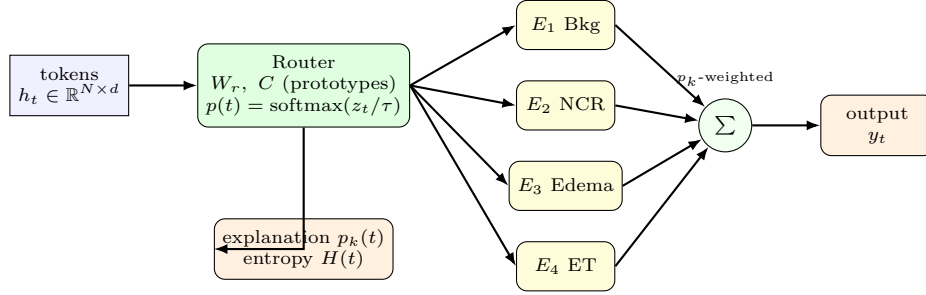


Fig. 2. The CCR bottleneck module. The router maps each token to a distribution $p(t)$ over the $K=4$ clinically-labeled experts via a prototype-augmented, temperature-scaled softmax (Eq. 1). Experts are combined by p -weighted soft dispatch during training and by hard arg max dispatch at inference (Eq. 2). The routing distribution itself, with its entropy, is emitted as the explanation.

3.3 The CCR bottleneck module

Figure 2 details the module. Given token h_t , the router computes concept logits by blending a learned linear projection with the cosine similarity to K learnable concept prototypes $C = [c_1, \dots, c_K]$, then a temperature-scaled softmax:

$$z_t = (1 - \beta) W_r h_t + \beta \text{sim}(h_t, C), \quad p(t) = \text{softmax}(z_t / \tau) \in \Delta^{K-1}, \quad (1)$$

where $W_r \in \mathbb{R}^{K \times d}$, $\text{sim}(h_t, C)_k = \frac{h_t^\top c_k}{\|h_t\| \|c_k\|}$, the blend β and temperature τ are (the latter annealed; Sec. 3.5). The prototypes give each concept an explicit anchor in feature space, stabilizing routing for rare concepts. Each expert E_k is a lightweight token-wise MLP. The dispatcher combines expert outputs differently at train and inference time:

$$\underbrace{y_t = \sum_{k=1}^K p_k(t) E_k(h_t)}_{\text{training (soft)}} \quad \underbrace{y_t = E_{k^*}(h_t), \quad k^* = \arg \max_k p_k(t)}_{\text{inference (hard)}}. \quad (2)$$

Soft dispatch keeps the routing differentiable so that the alignment loss can reshape the encoder; hard dispatch at inference makes the explanation a discrete, causal concept assignment—each region is processed by exactly one clinical expert. The module also emits the per-token entropy $H(t) = -\sum_k p_k(t) \log p_k(t)$ as a free uncertainty signal.

3.4 Training objective

The total loss has five terms. *Segmentation* $\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{focal}}$ supervises the decoder output against the labels. The key term is the *alignment* loss, a foreground-only focal binary cross-entropy that drives the routing toward the

Table 1. Three-phase training curriculum with temperature annealing. λ_1 and λ_4 are the alignment and boundary weights of Eq. 4.

Phase	Epochs	τ	λ_1 / λ_4	Purpose
Warm-up	1–10	2.0	0 / low	encoder learns features; routing unsupervised
Alignment	11–60	1.0	1 / 0.01	$\mathcal{L}_{\text{align}}$ shapes routing to concepts
Refinement	61–80	0.5	1 / high	sharpen assignment, emphasize boundaries

ground-truth concept maps. Let $m_k(t) \in \{0, 1\}$ be the token-level label for concept k (obtained by nearest-neighbour downsampling of the GT to the token grid) and $\text{FG} = \{t : m_0(t) = 0\}$ the foreground tokens. Then

$$\mathcal{L}_{\text{align}} = \frac{1}{|\text{FG}|(K-1)} \sum_{t \in \text{FG}} \sum_{k=1}^{K-1} w_k (1 - q_k(t))^\gamma \text{BCE}(p_k(t), m_k(t)), \quad (3)$$

where $q_k(t) = p_k(t)$ if $m_k(t) = 1$ else $1 - p_k(t)$, γ is the focal factor that down-weights easy tokens, and w_k re-weights rare concepts (NCR). Applying Eq. 3 on the simplex induces implicit competition: raising p_k on true- k tokens necessarily lowers the others. Restricting to foreground avoids the overwhelming background from drowning the rare tumor concepts. Three auxiliary terms regularize the routing: an *expert-diversity* loss $\mathcal{L}_{\text{div}} = \frac{1}{K(K-1)} \sum_{i \neq j} |\cos(r_i, r_j)|$ over the router’s per-expert directions r_k (discouraging redundant routing); an *entropy* term $\mathcal{L}_{\text{ent}} = \frac{1}{N} \sum_t H(t)$ (kept with a small weight $\lambda_3 \leq 0.01$, since larger values push routing toward uniform and destroy alignment); and a *boundary* term $\mathcal{L}_{\text{bnd}}$ that sharpens predicted edges. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1(\phi) \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{div}} + \lambda_3 \mathcal{L}_{\text{ent}} + \lambda_4(\phi) \mathcal{L}_{\text{bnd}}, \quad (4)$$

where λ_1, λ_4 depend on the curriculum phase ϕ .

3.5 Curriculum and temperature annealing

We introduce routing supervision gradually: warm-up delays $\mathcal{L}_{\text{align}}$ so the encoder settles before routing is forced, a precaution against early instability—though with a pretrained encoder it proves unnecessary in practice (A4, Sec. 5), so we retain it only as a safe default. We train in three phases with annealed temperature (Table 1). *Warm-up* ($\tau=2.0$, $\lambda_1=0$) lets the encoder learn discriminative features under segmentation loss alone. *Alignment* ($\tau=1.0$, $\lambda_1=1$) switches on $\mathcal{L}_{\text{align}}$ to shape the routing. *Refinement* ($\tau=0.5$) sharpens assignments and emphasizes boundaries. A lopsided expert utilization ($<5\%$ for any expert) after warm-up triggers re-initialization of that expert.

4 Faithfulness Framework

We separate three notions that the interpretability literature routinely conflates. *Faithfulness* asks whether the explanation reflects the model’s actual computa-

tion; *alignment quality* asks whether that explanation is clinically meaningful; and *causal relevance* asks whether the highlighted regions actually drive the prediction. Each requires its own evidence; only their conjunction supports the overall claim.

4.1 Definitions

Definition 1 (Concept Alignment Score). For concept k , $\text{CAS}(k) = \text{Pearson}(p_k(t), M_k(t))$, averaged over tokens t and validation cases, where $M_k(t)$ is the ground-truth concept- k indicator at token t . We report the foreground variant $\text{CAS}_{\text{fg}}(k)$ over $t \in \text{FG}$, $k \geq 1$; background is excluded because its near-constant mask makes the correlation ill-conditioned.

Definition 2 (Decoder Divergence). For a model with a separate decoder producing concept posterior $P_k^{\text{dec}}(t)$, $\text{DD}(k) = 1 - \text{Pearson}(P_k^{\text{dec}}(t), p_k(t))$ measures how far the decoder departs from the routing signal.

4.2 Structural faithfulness of CCR-Net

Proposition 1 (CCR-Net faithfulness). In CCR-Net the routing distribution $p(t)$ is the sole concept-assignment mechanism and is causally upstream of the prediction: the decoder receives, at the bottleneck, only the expert-processed features dispatched by p . The explanation $p(t)$ is therefore read directly from the forward computation rather than reconstructed from its output, so it incurs no post-hoc approximation gap. Its empirical alignment with clinical concepts is exactly CAS.

Argument. This is a structural statement about where the explanation comes from, not a quantitative theorem; we spell it out rather than leave it implicit. A post-hoc method A produces an explanation $A(f, x)$ as a function of the trained model f and input x , computed after $f(x)$; the map from $A(f, x)$ to the mechanism that produced $f(x)$ is approximate and method-dependent, which is the source of its unfaithfulness. In CCR-Net the quantity $p(t)$ is not a function of the output but an *intermediate state* of f itself: by Eqs. 1–2 every bottleneck feature passed to the decoder is $E_{k^*}(h_t)$ with $k^* = \arg \max_k p_k(t)$, so p determines which expert—hence which features—reach the decoder. The explanation is thus an identity on the computation, not an approximation of it; faithfulness in the structural sense (no reconstruction gap) holds by construction, and the only remaining empirical question—does this faithful explanation align with clinical ground truth—is answered by CAS. $\square$

Proposition 1 is structural: it establishes that the explanation is read from the forward computation with no post-hoc reconstruction gap, but it does *not* assert that the decoder output is a function of the routing alone. Two mechanisms let the prediction depend on more than p : training uses soft dispatch (all experts contribute), and the UNETR decoder receives multi-scale encoder skips that bypass the bottleneck. We quantify this residual with the Decoder Divergence

(Def. 2): on the test split $DD = 0.73, 0.53, 0.40$ for NCR, edema, and ET. DD is neither negligible—a decoder that merely echoed the 16^3 routing could not refine it to 128^3 boundaries—nor unstructured: it diverges most for the resolution-limited NCR and least for edema and ET, mirroring the CAS ordering. Structural faithfulness therefore does not by itself imply causal sufficiency, and we do *not* rest on a claim that the residual is small; we establish causal relevance directly, with deletion/insertion AUC on the inference-time *hard*-routing maps (Sec. 5.4).

4.3 Extending to arbitrary backbones

Conjecture 1 (Retrofit bound). Let the CCR module be inserted into a backbone whose decoder retains independent parameters, and let *explanation faithfulness* be operationalized as the measured deletion AUC of the hard-routing map. We conjecture that this quantity is bounded below by $CAS(k) \cdot (1 - DD(k))$.

Motivation and status. The intuition is that faithfulness should degrade in proportion to how far the decoder departs from the routing it is conditioned on—a departure measured by DD—scaled by how well the routing itself matches the clinical concepts, measured by CAS; the product is tight as $DD \rightarrow 0$. We state this as a conjecture rather than a theorem deliberately. It presumes that these two error sources compose multiplicatively and that deletion AUC is the right operationalization of faithfulness, and we prove neither; a derivation would require a formal model of how decoder skip pathways carry concept information around the bottleneck. What we can report is that the inequality *holds empirically* on every concept we measured (Fig. 4), which is evidence for the conjecture and not a substitute for a proof.

Although stated for the general setting, the bound applies to CCR-Net directly: its UNETR decoder is independently parameterized and skip-connected, and the measured DD above places CCR-Net in this regime rather than the idealized $DD=0$ limit. Combining that DD with CAS_{fg} (Table 2), the bound predicts a causal faithfulness of at least $CAS \cdot (1 - DD) = 0.18, 0.44, 0.56$ for NCR, edema, and ET; the deletion AUC we measure (0.68, 0.87, 0.82; Fig. 4) exceeds each, so the bound is satisfied empirically. A full retrofit study on a frozen third-party backbone is left to future work.

5 Experiments

5.1 Setup

We use the BraTS 2024 glioma training data [10,2]: four co-registered MRI modalities (T1, T1ce, T2, FLAIR) per case, with expert annotations remapped to $\{0:\text{Background}, 1:\text{NCR}, 2:\text{Edema}, 3:\text{ET}\}$. We split patient-wise 75/12.5/12.5% into train/validation/test ($n_{\text{test}}=168$). Inputs are z -scored per modality over nonzero voxels and processed at 128^3 with standard flips, rotations, and intensity augmentation. The encoder is initialized from self-supervised Swin-UNETR

Table 2. Segmentation, alignment, AUROC, and Decoder Divergence (DD) on the BraTS test split ($n=168$). CAS_{fg} is the foreground concept-alignment score; lower DD means the decoder tracks the routing more closely.

	WT / NCR	TC / Edema	ET
Dice	0.915	0.845	0.837
CAS_{fg} (NCR/Edema/ET)	0.688	0.935	0.930
AUROC (NCR/Edema/ET)	0.711	0.832	0.774
DD (NCR/Edema/ET)	0.733	0.531	0.395

weights. We train for 80 epochs with AdamW ($\text{lr}=10^{-4}$, cosine schedule), mixed precision, and the curriculum of Table 1. We report Dice and HD95 for whole tumor (WT), tumor core (TC) and enhancing tumor (ET); CAS_{fg} and one-vs-rest AUROC per concept; deletion/insertion AUC for causal relevance; and, for every explanation method, concept-discriminability AUROC (Sec. 5.4).

5.2 Metrics for faithfulness

Given a per-voxel explanation e_k for concept k , we rank voxels by e_k and measure the concept- k confidence $s_k(\cdot) = \text{mean}_{v \in M_k} \text{softmax}_k$ over the ground-truth region M_k . *Deletion* progressively zeroes the top fraction τ of voxels and records the normalized confidence *removed*; *insertion* reveals the top τ from a blank volume and records the confidence *recovered*:

$$\text{Del-AUC} = \int_0^1 \left(1 - \frac{s_k(\tau)}{s_k(0)}\right) d\tau, \quad \text{Ins-AUC} = \int_0^1 \frac{s_k(\tau)}{s_k(1)} d\tau, \quad (5)$$

both in $[0, 1]$, higher meaning stronger causal relevance (the top-ranked voxels drive the prediction). Crucially, we apply the *identical* protocol of Eq. 5 to CCR routing and to every baseline, so the comparison isolates the explanation, not the scoring.

5.3 Segmentation and alignment

Table 2 reports the main results. CCR-Net segments competitively (Dice WT 0.915) while producing routing that aligns strongly with the clinical concepts: CAS_{fg} reaches 0.94 (edema) and 0.93 (ET). Necrotic core is the weak concept (0.68): at the 16^3 bottleneck a case contains only tens of NCR tokens, so the Pearson correlation is intrinsically noisy—an honest resolution limit, not a model failure, consistent across Dice, AUROC, and CAS.

5.4 Faithfulness vs. post-hoc attribution

Faithfulness is not an artifact of the scoring window. Because the alignment loss trains the routing to sit on the ground-truth region, scoring deletion over

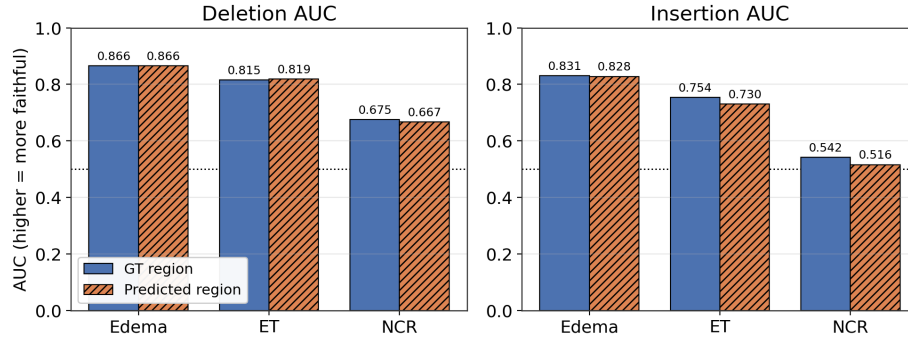


Fig. 3. CCR faithfulness is window-invariant. Deletion (left) and insertion (right) AUC scored over the ground-truth region (GT) vs. the model’s own predicted region are near-identical—deletion differs by ≤ 0.008 —ruling out a supervised-localization confound; dotted line marks the 0.5 reference. Per-cell n (GT/pred): edema 168/168, ET 163/168, NCR 67/91.

that same region could in principle reward supervised localization rather than faithfulness. We rule this out by scoring in *both* windows: over the ground-truth region M_k and over the model’s own *predicted* region ($\arg \max_k$), which carries no ground-truth supervision. Figure 3 shows the two are near-identical—deletion differs by at most 0.008 and insertion by at most 0.026—so the measured faithfulness is a property of the routing, not of the scoring window. Necrotic core, moreover, is now measured on 67–91 cases rather than a handful, removing the small-sample caveat of earlier drafts.

We compare CCR routing, on the same trained network and under the identical protocol of Eq. 5, against two families: four post-hoc baselines (Grad-CAM, gradient saliency, integrated gradients (IG), occlusion) and—critically—the network’s own **segmentation posterior** (softmax_k of the seg head, denoted SEG-POST), obtained free from the same forward pass. SEGPOST is the baseline that concept-explanation evaluations typically omit, and omitting it would make our comparison unfair: a segmentation head is trained on precisely these concept labels, so it is the strongest possible concept discriminator. Table 3 reports all three axes on all 168 test cases.

Sensitivity: CCR does not win it, and the winners are informative. On deletion the sensitivity methods score highest—IG and occlusion reach 0.95, above SEG-POST’s 0.91 and CCR’s 0.79. This is expected: deletion rewards concentrating importance on the few most output-sensitive voxels, which IG (by its completeness axiom) and occlusion (whose saliency *is* a coarse deletion) optimize directly. SEGPOST scores high for a different and more revealing reason—ranking voxels by the model’s own confidence and then measuring that same confidence is a closed loop, so its deletion score is close to tautological. Deletion is a necessary check, and CCR clears it far above the 0.5 floor, above Grad-CAM and gradient,

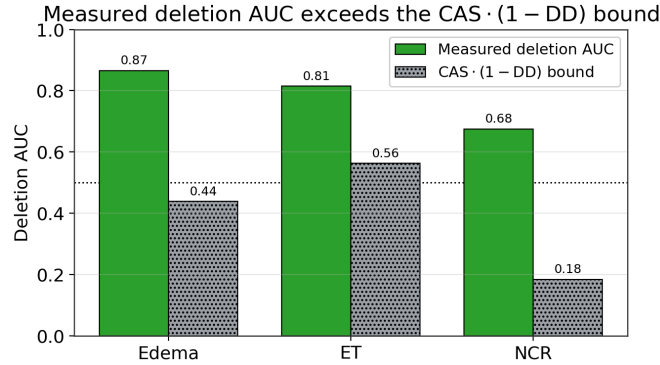


Fig. 4. The measured deletion AUC (green) exceeds the $\text{CAS} \cdot (1 - \text{DD})$ lower bound (gray) of Proposition 1 for every concept, confirming the bound empirically; dotted line marks the 0.5 reference.

and window-invariantly (Fig. 3); but it does not identify a *concept* explanation, and we do not claim it.

Insertion breaks the pattern, and the break is diagnostic. The self-reference that lifts SEGPOST on deletion does not transfer: **CCR exceeds SegPost on insertion** (0.709 vs. 0.680 mean), significantly so on necrotic core (+0.133 [0.087, 0.178], $p < 0.001$; +0.130 on the predicted window), tying on edema ($p=0.31$) and losing modestly on ET (−0.046). Recovering a prediction requires restoring coherent predictive *context*, not merely the voxels the model was already confident about—and on the hard class the posterior’s voxel-level ranking is unreliable where the routing’s regional assignment is not. This is direct evidence that routing carries information the output does not, rather than a degraded copy of it.

Concept-discriminability, and the ceiling that limits its interpretation. A clinical explanation must say *which* concept a region is. We measure this directly: for each method and concept k we compute the AUROC of its map e_k as a classifier for GT concept k among the other tumor voxels ($\text{GT} > 0$, $\text{GT} \neq k$). Against the post-hoc family the result is decisive: CCR reaches 0.85–0.89 across all three concepts while every post-hoc method is far weaker—occlusion 0.64–0.80, gradient 0.46–0.70, IG 0.49–0.64 (chance on edema), Grad-CAM ≈ 0.5 (chance throughout). Pooled, CCR scores 0.88 [0.87, 0.89] against occlusion’s 0.73 [0.71, 0.74], a gap of +0.15 with non-overlapping bootstrap CIs and paired Wilcoxon $p < 0.001$ against every baseline. The methods that win deletion cannot tell the concepts apart: their per-class maps highlight the same high-signal tumor structure regardless of k (Fig. 6).

SEGPOST, however, scores 0.98—above CCR. We report this plainly, and we argue it shows the metric saturates rather than that the posterior explains better. Two observations support this. First, SEGPOST reaches 0.982 on CCR-Net and

0.979 on CCR-SegResNet (Sec. 5.6) despite different backbones, different pre-training, and different Dice: a score that is invariant to everything except being a competent segmenter is measuring segmentation competence, not explanation quality. Under this window the posterior’s AUROC is close to a restatement of its own accuracy, since ranking voxels is far easier than thresholding them. Second, and decisively for our criteria, SEGPOST *is* the prediction: its decoder divergence is 0 by construction, so it satisfies concept-labeling only by forfeiting upstream-ness entirely. An explanation identical to the thing explained answers no question about how the output arose.

How much of the residual gap is decision budget? CCR routing decides over $16^3=4096$ tokens; the posterior decides over 128^3 voxels— $512\times$ more decision units. The comparison above therefore confounds information with spatial resolution. We isolate the two with a granularity-matched control, SEGPOST₁₆: the posterior average-pooled onto the router’s own token grid and restored through the *identical* trilinear operator used for routing, so both maps carry exactly the same spatial degrees of freedom. Matching granularity removes roughly half the gap—the paired CCR deficit narrows from -0.106 $[-0.116, -0.096]$ to -0.051 $[-0.059, -0.043]$ on CCR-Net, and from -0.090 to -0.039 on CCR-SegResNet (52% and 57% of the gap respectively). Per concept the picture is sharper: on *edema* the two become statistically indistinguishable on both backbones (paired difference -0.003 $[-0.014, +0.008]$ on Swin, $+0.007$ $[-0.004, +0.019]$, $p=0.70$, on the CNN, with CCR nominally ahead), i.e. at equal decision budget the bottleneck loses *nothing but resolution* for that concept. Necrotic core and ET retain a genuine gap (-0.07 to -0.10 , $p < 0.001$), concentrated in exactly the concepts the 16^3 bottleneck under-resolves. We emphasize that this control does not restore a discriminability win—SEGPOST₁₆ still exceeds CCR overall—it quantifies how much of the apparent deficit is architectural budget rather than lost concept information.

5.5 Qualitative explanation

Figure 7 shows the routing read-out for a representative case. The hard routing assignment (panel 4) reproduces the ground-truth and predicted segmentation *within the tumor*—the routing tracks the prediction (though not identically; DD)—while the per-concept soft maps localize each clinical concept to its sub-region (NCR in the core, edema on the rim, ET in the enhancing ring). A clinician reads these maps directly as {NCR, Edema, ET}, which a gradient heatmap cannot provide. In a representative hard case (Fig. 9) the routing flags enhancing tumor that the decoder under-segments, showing that the routing explanation can carry diagnostic signal independent of—and more informative than—the final mask.

5.6 CCR is backbone-agnostic: a CNN instantiation

To test whether the CCR principle is specific to the Swin transformer, we drop the *same* module—identical router, experts, five-term objective, and curriculum—

Table 3. Three axes on the BraTS test split ($n=168$, identical protocol, per-concept mean, ground-truth window). *Deletion* and *insertion* measure input sensitivity; *concept-discriminability* is concept k vs. the other tumor voxels (0.5=chance). The lower block is the model’s own output, not a post-hoc explanation: it is reported because omitting it would overstate our case, but it *is* the prediction (DD=0), so its discriminability is not evidence of explanatory power. Best post-hoc value per column in **bold**; *Upstream?* marks whether the map is causally prior to the prediction, *Concepts?* whether it can separate them above chance.

Method	Deletion	Insertion	Concept-discrim.	Upstream?	Concepts?
<i>Post-hoc saliency</i>					
Grad-CAM	0.416	0.358	0.521	✓	✗
Gradient saliency	0.640	0.449	0.601	✓	✗
Integrated gradients	0.952	0.909	0.583	✓	✗
Occlusion	0.952	0.929	0.741	✓	✗
<i>Intrinsic, upstream of the prediction</i>					
CCR (ours)	0.785	0.709	0.872	✓	✓
<i>The model’s own output — not an explanation</i>					
SEGPOST	0.912	0.680	0.982	✗	✓
SEGPOST ₁₆ (granularity-matched)	—	—	0.933	✗	✓

into a SegResNet CNN encoder–decoder (**CCR-SegResNet**), replacing only the backbone; the module sits unchanged at the CNN’s 192×16^3 bottleneck. Trained *from scratch* (no self-supervised pretraining, unlike CCR-Net), it reproduces every headline result (Table 4): concept alignment (CAS_{fg} 0.80/0.96/0.95), concept-discriminability (0.85/0.91/0.89, again far above the best post-hoc baseline’s 0.74), and competitive segmentation; its CAS follows the *identical* training dynamics—flat under warm-up, then rising sharply the moment $\mathcal{L}_{\text{align}}$ activates. CCR is therefore a *backbone-agnostic*, drop-in intrinsic-explanation module, not a transformer-specific model. We do not read the CNN’s marginally higher Dice as CNNs beating transformers—SegResNet is a strong BraTS backbone—the point is that the principle transfers; its hard-routing utilization is more skewed than the transformer’s, which lowers whole-volume AUROC but not the foreground concept metrics reported.

The second backbone also supplies the control for our reading of the discriminability metric in Sec. 5.4. SEGPOST scores 0.982 on CCR-Net and 0.979 on CCR-SegResNet, and SEGPOST₁₆ scores 0.931 and 0.933 respectively—near-identical across two architectures that differ in backbone family, pretraining regime, and Dice. A metric that returns the same value for any competent segmenter is characterizing the segmenter, not the explanation. The granularity-matched edema tie likewise replicates on both backbones, which is why we read it as a property of the 16^3 bottleneck rather than a quirk of one run.

We note explicitly what this section does *not* establish. We did not train plain Swin-UNETR or plain SegResNet counterparts under our split, so we make no claim that the CCR bottleneck is accuracy-neutral; the Dice values here are in

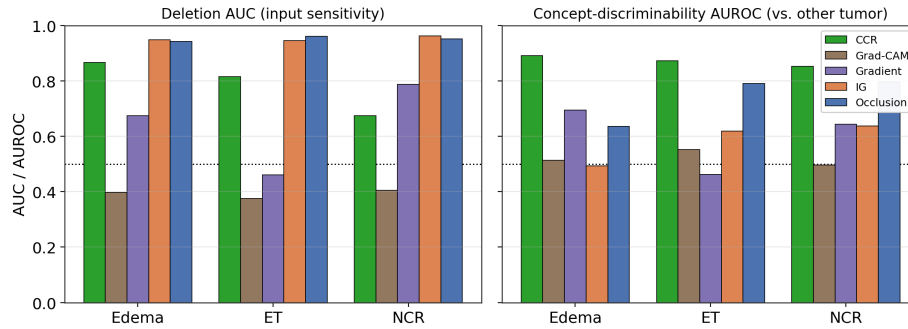


Fig. 5. No single method leads on both axes (per concept, BraTS test). *Left:* deletion AUC (input sensitivity)—integrated gradients and occlusion dominate. *Right:* concept-discriminability AUROC (concept vs. other tumor; dotted line = 0.5 chance)—the sensitivity methods fall toward chance while CCR rises far above them. The segmentation posterior (Table 3) would sit at the top of the right axis, but it is the prediction itself rather than an explanation of it. Among maps that are genuinely *up-stream* of the prediction, CCR is alone in the upper region of the right axis.

Table 4. CCR is backbone-agnostic. The same module in a Swin transformer (CCR-Net) and a SegResNet CNN (CCR-SegResNet), BraTS test ($n=168$). The CNN trains from scratch yet reproduces the concept alignment, concept-discriminability, and segmentation of the pretrained transformer.

Instantiation	Dice WT/TC/ET	CAS _{fg} N/E/T	Concept-discrim.	SEGPST / SEGPST ₁₆
CCR-Net (Swin, pretrained)	0.915/0.845/0.837	0.69/0.94/0.93	0.872	0.982 / 0.931
CCR-SegResNet (CNN, scratch)	0.927/0.883/0.869	0.80/0.96/0.95	0.883	0.979 / 0.933

line with published results for these backbones on BraTS but are contextual rather than controlled comparisons (see Sec. 6).

5.7 Ablations

We ablate two training choices: the alignment loss and the warm-up phase. **A1 (no $\mathcal{L}_{\text{align}}$)** disables the alignment loss in every phase; **A4 (no warm-up)** activates alignment from epoch 1. Table 5 reports the outcome—one choice is essential, the other is not.

A1 confirms the first hypothesis: without $\mathcal{L}_{\text{align}}$, CAS_{fg} collapses to near zero for NCR and ET (-0.01 , 0.09) and to 0.32 for edema—AUROC falls to chance or below—while Dice is essentially preserved (WT 0.905 vs. 0.915). Concept alignment is therefore *learned* from $\mathcal{L}_{\text{align}}$, not an incidental by-product of segmentation—exactly the temporal signal in Fig. 8. Crucially, the collapse reaches the headline metric: A1’s concept-discriminability falls from 0.87 to 0.61 (NCR to chance), and CCR’s routing then no longer exceeds the best post-hoc baseline (occlusion 0.74)— $\mathcal{L}_{\text{align}}$ is precisely what makes the routing a *concept-discriminative*

Table 5. Ablations on the test split (run to 40 epochs; the effect is evident well before convergence). A1 (no alignment loss) collapses CAS while preserving Dice—alignment is *learned*. A4 (no warm-up) matches the full model on CAS and Dice with only a mild utilization skew—the design is robust and warm-up is a stabilizing default, not a requirement.

	Full CCR-Net A1: no $\mathcal{L}_{\text{align}}$ A4: no warm-up		
CAS _{fg} (Edema)	0.935	0.322	0.932
CAS _{fg} (ET)	0.930	0.094	0.927
CAS _{fg} (NCR)	0.688	−0.011	0.666
Dice (WT)	0.915	0.905	0.908
Max expert utilization (%)	28	32	41

explanation rather than a sensitivity map. A4 tells the opposite, and equally informative, story: activating $\mathcal{L}_{\text{align}}$ from epoch 1 leaves CAS and Dice essentially unchanged (CAS_{fg} 0.93/0.93/0.67; Dice WT 0.908) with *no* expert collapse. The pretrained encoder, prototype-anchored router, and always-on diversity loss make warm-up a stabilizing default rather than a requirement; its only measurable effect is more balanced utilization (max 28% with warm-up vs. 41% without). We retain warm-up as a safe default but do not depend on it.

5.8 Training dynamics

Figure 8 traces CAS_{fg} over training. Alignment is near zero during warm-up (routing unsupervised), rises sharply the moment $\mathcal{L}_{\text{align}}$ activates at epoch 11, and is held through refinement; edema and ET converge near 0.93 while NCR plateaus at the resolution-imposed ceiling. The curve is the temporal mirror of the A1 ablation: removing $\mathcal{L}_{\text{align}}$ removes exactly the rise visible at epoch 11.

5.9 CCR-Retrofit (supplementary)

Proposition 1 predicts that the CCR module transfers to a *frozen*, independently-trained backbone—training only the router and experts while the encoder and decoder weights are held fixed—with faithfulness bounded below by $\text{CAS} \cdot (1 - \text{DD})$. This *CCR-Retrofit* setting tests whether the principle carries to a backbone the method did not design. We freeze a pretrained Swin-UNETR, insert the CCR module at the stage-2 bottleneck, train only the CCR parameters under $\mathcal{L}_{\text{align}}$, and report CAS, DD, and deletion/insertion AUC, verifying the bound. [Optional supplementary experiment—results pending, included if space permits. A frozen decoder cannot adapt to the routing, so we expect a larger DD than end-to-end CCR-Net while faithfulness continues to respect the $\text{CAS} \cdot (1 - \text{DD})$ bound.]

6 Discussion and Limitations

No controlled accuracy comparison. We did not train plain-backbone counterparts (Swin-UNETR or SegResNet without the CCR bottleneck) under our data split. We therefore make *no* claim that CCR is accuracy-neutral: our Dice results show that CCR-Net and CCR-SegResNet segment competitively and in line with published values for these backbones, but the cost—if any—of inserting the routing bottleneck is unquantified here. This is the most direct experiment a follow-up should run, and the ablation configurations are released with the code.

Necrotic core resolution. At the 16^3 bottleneck a case contributes only tens of NCR tokens, capping CAS and faithfulness for that concept; this is the single consistent weak spot across all metrics, and it is the same concept where the granularity-matched control leaves a residual gap to the segmentation posterior. A finer bottleneck trades semantic context for resolution, and reconciling the two is open.

What the metrics can and cannot settle. Deletion, insertion, and concept-discriminability were all designed to evaluate post-hoc saliency maps: they ask whether a map has found the influential voxels. CCR’s claim is different in kind—that there is no gap between the explanation and the computation, because the routing *is* the decision. We report these metrics because a reader is entitled to ask whether the routing is causally real, and they answer that (window-invariant deletion well above chance, insertion above the posterior’s on the hard class). But we caution against reading the leaderboard as the contribution: the methods that top the sensitivity axes cannot label a concept at all, and the method that tops discriminability is the prediction itself. CCR is not first on any single column of Table 3; it is the only entry that is concept-labeled *and* upstream of the output, which is the conjunction a clinical explanation requires.

Calibration. Routing entropy provides a free per-voxel uncertainty map, but its absolute calibration is poor (expected calibration error ≈ 0.73): peaked routing at low temperature is confident even where token-level accuracy is low. We therefore treat entropy as a qualitative uncertainty signal and make no calibration claim.

Soft training vs. hard inference. During training all experts contribute via soft weighting; the faithfulness claim concerns inference-time hard dispatch, and our deletion/insertion analysis uses hard-routing maps accordingly (Sec. 5.4).

Concept granularity. $K=4$ imposes hard concept boundaries; transition-zone tokens are forced into one concept, matching the BraTS protocol but discarding mixed-tissue nuance.

Scope of the explanation. CCR explains at the level of clinical *concepts*—which of the K experts processed each region, and with what confidence—not a finer account of the router’s internal reasoning, and the concept content necessarily overlaps the segmentation it conditions. CCR is thus best understood as an inherently interpretable segmentation model whose routing decision is a faithful, intrinsic explanation, rather than a post-hoc saliency method: its advantage over a heatmap is faithfulness, concept labels, modularity, and free uncertainty—not spatial sharpness. Consistently, strong sensitivity methods (integrated gradients, occlusion) score higher on deletion—an input-sensitivity metric they optimize directly—while failing to separate the clinical concepts, and the segmentation posterior separates concepts better while

being the output itself (Sec. 5.4); we report both openly rather than claiming to win every metric.

7 Conclusion

We introduced Clinical Concept Routing: making the routing decision at the segmentation bottleneck the faithful, clinically-labeled, per-voxel explanation, computed in one forward pass. We detailed the router, expert bank, dispatch, objective, and curriculum that make this work, and characterized the sense in which CCR-Net’s explanation is structurally faithful, with alignment quality measured by the Concept Alignment Score. Evaluating against post-hoc saliency *and* against the model’s own segmentation posterior under one identical protocol, we find that no method leads everywhere: integrated gradients and occlusion win the sensitivity axes they optimize but cannot separate the clinical concepts, while the posterior separates them almost perfectly and yet *is* the prediction, with zero decoder divergence. CCR routing is the only map that is concept-discriminative (0.88, far above the best post-hoc method) while remaining causally upstream of the output—and it exceeds the posterior on insertion, evidence that it carries information the output does not. Matching the posterior to the router’s decision budget removes about half the residual discriminability gap and closes it entirely on edema. The principle is *backbone-agnostic*: the same module, dropped unchanged into a SegResNet CNN (CCR-SegResNet), reproduces the concept alignment, discriminability, and segmentation from scratch. CCR delivers intrinsic, concept-labeled explanation for 3D volumetric dense prediction, where prior intrinsic methods address only scalar-output classification. Because CCR-Net’s decoder is independently parameterized and skip-connected, its measured Decoder Divergence already places it in the retrofit regime of Proposition 1: the retrofit bound is thus empirically *supported* (Fig. 4), not merely asserted. A full CCR-Retrofit study on a frozen third-party backbone (Sec. 5.9) and extension to additional anatomies are promising future work.

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Per-concept explanation maps (case BraTS-GLI-02284-100, slice 71) — cyan = GT concept boundary

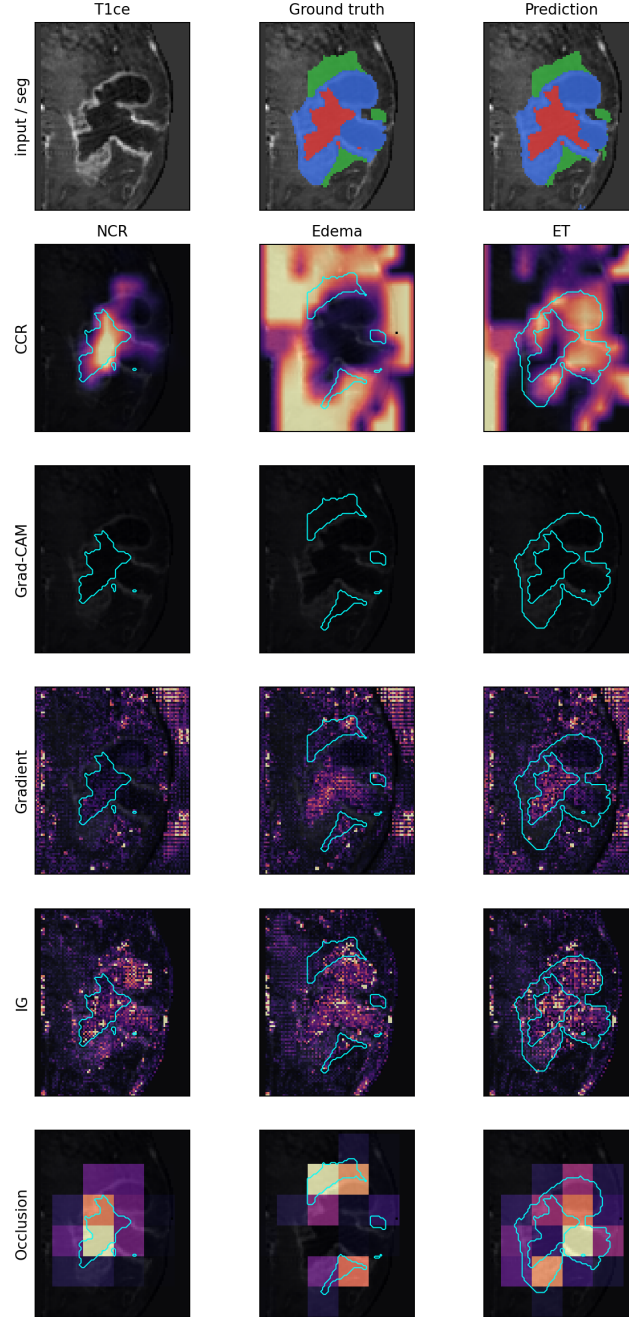


Fig. 6. Per-concept explanation maps (case BraTS-GLI-02284, one axial slice; cyan = GT concept boundary). *Rows*: input/segmentation, then each method; *columns*: NCR, edema, ET. CCR’s three maps *differ* and concentrate inside the correct boundaries—a concept partition—whereas Grad-CAM, gradient, and IG produce nearly the *same* map for every concept (high-sensitivity structure, not a concept assignment) and occlusion is coarse and only partly concept-specific. This is the qualitative face of Table 3 (right).

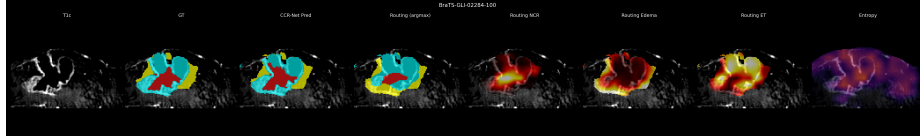


Fig. 7. Routing maps (case BraTS-GLI-02284). Left to right: T1ce, ground truth, prediction, hard routing assignment, soft routing for NCR / Edema / ET, and routing entropy. The argmax routing matches GT/prediction inside the tumor; each soft map localizes its concept; entropy peaks at concept boundaries.

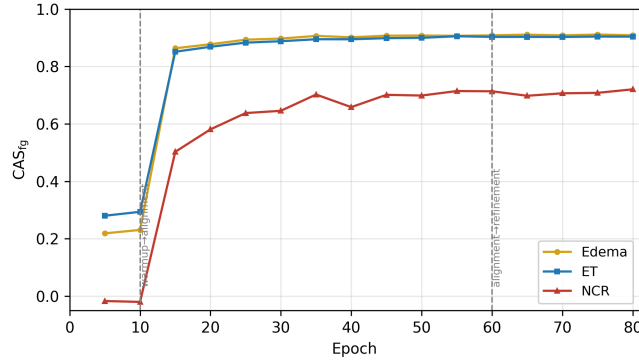


Fig. 8. CAS_{fg} across the three-phase curriculum (dashed lines: phase boundaries). Alignment emerges only after $\mathcal{L}_{align}$ activates at epoch 11, directly mirroring the A1 ablation.

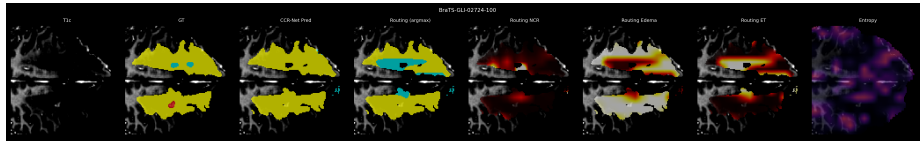


Fig. 9. Hard case (BraTS-GLI-02724): the routing assignment flags enhancing tumor that the final decoder mask under-segments, illustrating that the routing explanation can be more informative than the prediction itself.